Chapter 9 • Similar Figures

Solved Exercise 9.3

1. Sphere radii are in ratio 3:4. Find volume ratio.
$3^3:4^3=27:64$.

2. Regular tetrahedron volumes are in ratio 8:27. Find side ratio.
$\sqrt[3]{8}:\sqrt[3]{27}=2:3$.

3. Right-cone volumes are in ratio 64:125. Find height and base-area ratios.
(i) Heights = $\sqrt[3]{64}:\sqrt[3]{125}$
= 4:5
(ii) base areas = $4^2:5^2$
= 16:25.

4. Find the four missing measures in the similar-solid diagrams.
(i) $V_1 = 1536(12/16)^3$
$= 1536(\frac{3}{4})^3$
$= 648\ cm^3$.
(ii) $V_1 = 171.5(2.5/8.75)^3$
$= 171.5(\frac{2}{7})^3$
$= 4\ cm^3$.
(iii) $\frac{A_1}{A_2} = \frac{392}{162}$
$= \frac{196}{81}$
$= (\frac{14}{9})^2$, so V_1
$= 729(\frac{14}{9})^3$
$= 2744\ cm^3$.
(iv) $\frac{V_1}{V_2} = \frac{64}{216}$
$= \frac{8}{27}$
$= (\frac{2}{3})^3$, so r_1
$= 12(\frac{2}{3})$
$= 8$ cm.

5. Two similar cans have corresponding lengths in ratio 3:2 (larger:smaller).
(i) Smaller surface area = $96(2/3)^2$
$= \frac{128}{3} \approx 42.67\ m^2$.
(ii) Larger volume = $240(3/2)^3$
$= 810\ m^3$.

6. Two similar tanks have heights in ratio 5:3 (larger:smaller).
(i) Smaller surface area = $250(3/5)^2$
$= 90\ m^2$.
(ii) Larger volume = $270(5/3)^3$
$= 1250\ m^3$.

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